CPEN 416

# Lecture 13: QFT and quantum phase estimation

Tuesday 27 October 2026

# Announcements

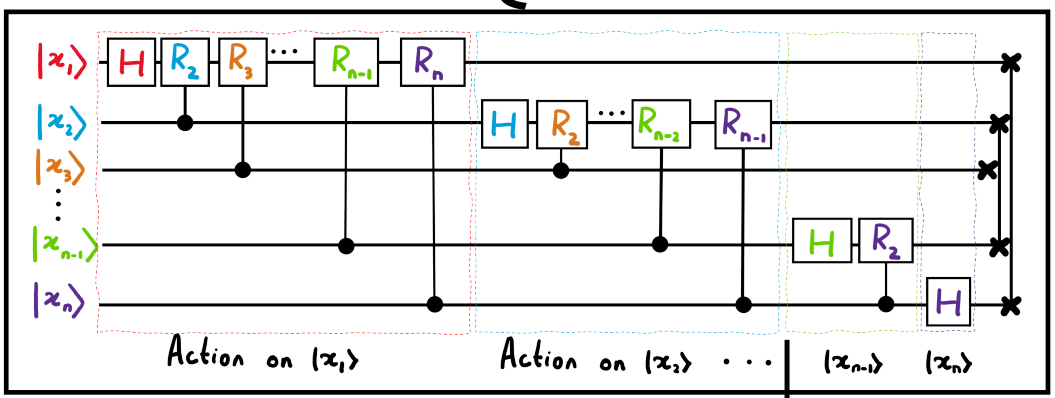
preskill

- Quiz 6 today
- Tutorial Assignment 2 due Friday 23:59
- Project midterm checkpoint due Monday; sign up for appointment time in Canvas ~~tomorrow~~ [available now]

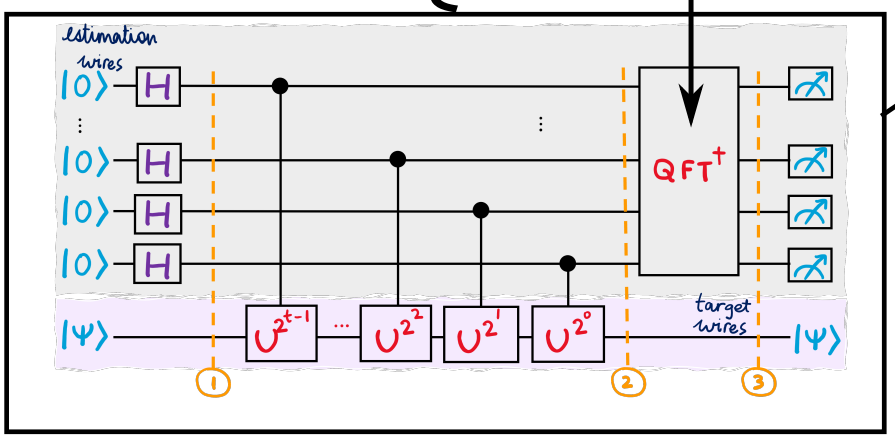
*indicate group ID if possible

# Reminder: where are we going?

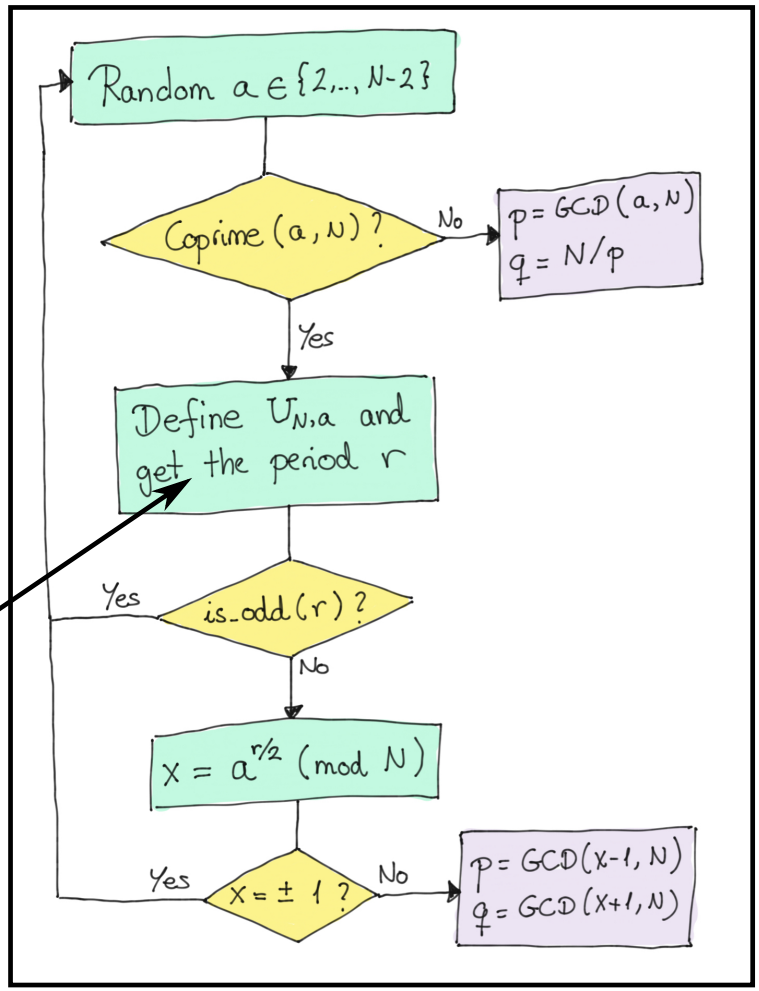
## 1. QFT




## 2. QPE




## 3. Shor



# Last time

We introduced the quantum Fourier transform, which is analogous to the inverse discrete Fourier transform.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

$$QFT = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)(N-1)} \end{pmatrix}$$

where for $n$ qubits, $N = 2^n$, and $\omega = e^{2\pi i/N}$

# Learning outcomes

- Describe the behaviour of the quantum Fourier transform
- Construct a circuit for the quantum Fourier transform and analyze its resource usage
- Outline the algorithmic steps of the QPE subroutine

# Quantum Fourier transform: 1 qubit

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

3-qubit case: $|k=3\rangle = |011\rangle$

$N = 2^1 = 2$

$\omega = e^{\frac{2\pi i}{2}} = e^{\pi i} = -1$

$$\Rightarrow QFT_1|x_1\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^{1} (-1)^{x_1 \cdot k} |k\rangle$$

$$= \frac{1}{\sqrt{2}} \left[ (-1)^{x_1 \cdot 0} |0\rangle + (-1)^{x_1 \cdot 1} |1\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[ |0\rangle + (-1)^{x_1} |1\rangle \right]$$

$e^{\frac{2\pi i}{2} \cdot x_1} = e^{2\pi i \cdot \frac{x_1}{2}}$

$0.x_1$

$QFT|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$

$QFT|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

# Quantum Fourier transform: 2 qubits

$e^{2\pi i} = 1$

$e^{\frac{2\pi i}{4}} = e^{\frac{\pi i}{2}} = i$

$\left(e^{\frac{2\pi i}{4}}\right)^4 = 1$

$\omega^5 = \omega \quad \omega^6 = \omega^2 \ldots$

$$QFT = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{pmatrix}$$

0 1 2 3
4 5 6 7
8 9 10 11
12
⋮

1 $\omega$ $\omega^2$ $\omega^3$

Re-express higher powers of $\omega = e^{\frac{2\pi i}{4}}$:

$\omega^4$ $\omega^6$
$\omega^6$ $\omega^9$

Initially rewrite as:

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega^1 \end{pmatrix}$$

Simplify as:

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

gate guess: an S, two Hadamards, some two-qubit gate.

# Quantum Fourier transform: 2 qubits

$$QFT = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega^1 \end{pmatrix}$$

How does this affect the computational basis states?

$QFT|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

$QFT|01\rangle = \frac{1}{2}(|00\rangle + \omega^{(i)}|01\rangle + \omega^{2(i)}|10\rangle + \omega^{3(i)}|11\rangle)$

$QFT|10\rangle = \frac{1}{2}(|00\rangle + \omega^2|01\rangle + |10\rangle + \omega^2|11\rangle)$

$QFT|11\rangle = \frac{1}{2}(|00\rangle + \omega^3|01\rangle + \omega^2|10\rangle + \omega|11\rangle)$

# Quantum Fourier transform: 2 qubits

$$QFT|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$QFT|01\rangle = \frac{1}{2}(|00\rangle + w^{(i)}|01\rangle + w^{(-1)}|10\rangle + w^{(-i)}|11\rangle)$$

$$QFT|10\rangle = \frac{1}{2}(|00\rangle + w^2|01\rangle + |10\rangle + w^2|11\rangle)$$

$$QFT|11\rangle = \frac{1}{2}(|00\rangle + w^3|01\rangle + w^2|10\rangle + w|11\rangle)$$

$$QFT|00\rangle = \frac{1}{2}(|0\rangle+|1\rangle)\otimes(|0\rangle+|1\rangle) = (H\otimes H)|00\rangle$$

$$QFT|01\rangle = \frac{1}{2}\left((|0\rangle+w^2|1\rangle)\otimes|0\rangle + (w|0\rangle + w^3|1\rangle)\otimes|1\rangle\right)$$

$$= \frac{1}{2}\left((|0\rangle+w^2|1\rangle)\otimes(|0\rangle+w|1\rangle)\right)$$

$$QFT|10\rangle = \frac{1}{2}\left((|0\rangle+|1\rangle)|0\rangle + w^2(|0\rangle+|1\rangle)\otimes|1\rangle\right)$$

$$= \frac{1}{2}\left((|0\rangle+|1\rangle)\otimes(|0\rangle+w^2|1\rangle)\right)$$

$$QFT|11\rangle = \frac{1}{2}\left((|0\rangle+w^2|1\rangle)\otimes(|0\rangle+w^3|1\rangle)\right)$$

# Quantum Fourier transform: 2 qubits

$$QFT\,|x_1x_2\rangle = \frac{1}{2}\left[(|0\rangle + \omega^{2x_2}|1\rangle)\otimes(|0\rangle + \omega^{2x_1+x_2}|1\rangle)\right]$$

0. ..

$$= \frac{1}{2}\left[\left(|0\rangle + e^{2\pi i\cdot\frac{2x_2}{4}}|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i\cdot\frac{(2x_1+x_2)}{4}}|1\rangle\right)\right]$$

$$= \frac{1}{2}\left[\left(|0\rangle + e^{2\pi i\cdot 0.x_2}|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i\cdot 0.x_1x_2}|1\rangle\right)\right]$$

$$\frac{x_1}{2} + \frac{x_2}{4} = 0.x_1x_2$$

3 qubit guess:

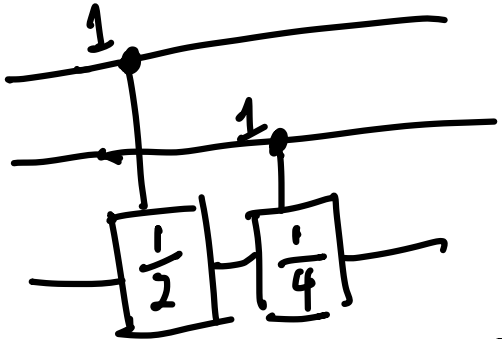
$$\frac{1}{2\sqrt{2}}\left(\left(|0\rangle + e^{2\pi i 0.x_3}|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i\,0.x_2x_3}|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i\,0.x_1x_2x_3}|1\rangle\right)\right.$$

# Quantum Fourier transform: 3 qubits

Define $R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{pmatrix}$

$$QFT = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & \cdots \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 & \cdots \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \cdots \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \cdots \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \cdots \end{pmatrix}$$

$$QFT|x_1x_2x_3\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_3}|1\rangle\right) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_2x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle)$$




↳ intuition

$e^{2\pi i \left[\frac{x_1}{2} + \frac{x_2}{4} + \frac{x_3}{8}\right]}$

# Quantum Fourier transform: $n$ qubits

General expression:

$$\begin{aligned} |x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\ &= \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}} \end{aligned}$$

Full mathematical derivation at end of slides.

# A circuit for the QFT

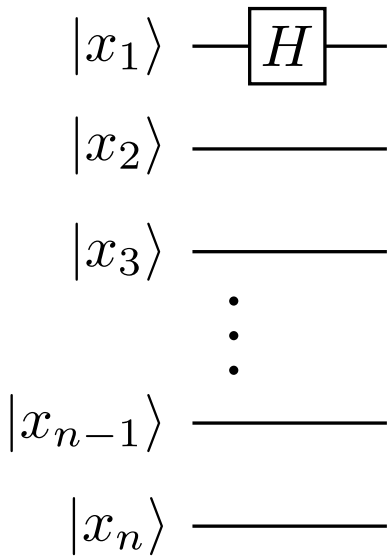
$$|x\rangle \to \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

Starting with the state

$$|x\rangle = |x_1 \cdots x_n\rangle,$$

apply a Hadamard to qubit 1:

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1}|1\rangle\right)|x_2\cdots x_n\rangle$$

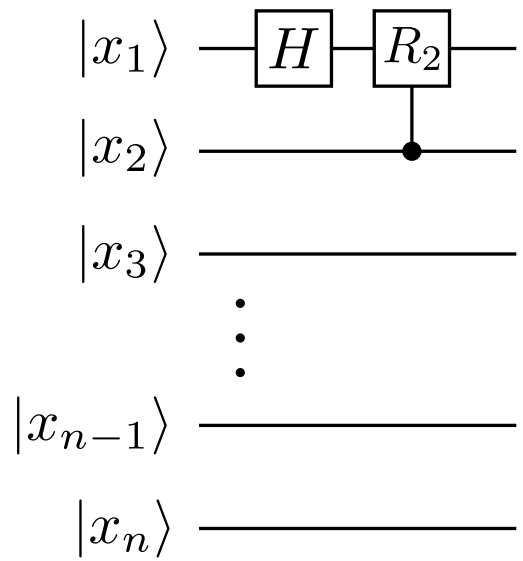


# A circuit for the QFT

$$|x\rangle \to \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

Apply controlled $R_2$ from qubit $2 \to 1$

$$R_2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^2}} \end{pmatrix}$$




$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1}|1\rangle\right)|x_2 \cdots x_n\rangle$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1x_2}|1\rangle\right)|x_2 \cdots x_n\rangle$$

# A circuit for the QFT

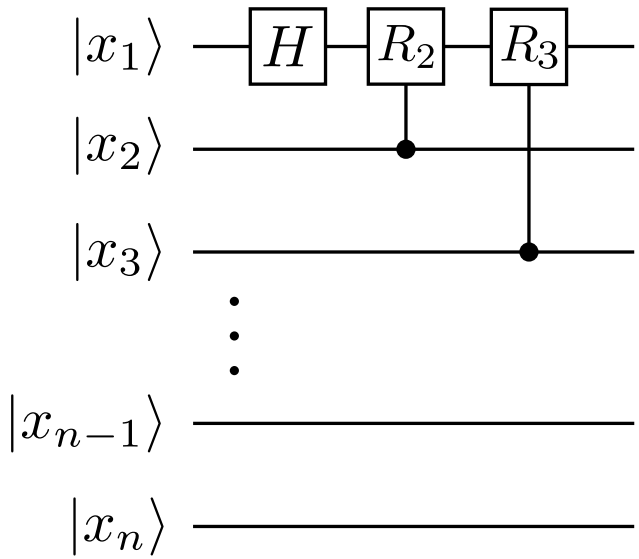
$$|x\rangle \to \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

$0.x_1x_2x_3$

$\frac{x_1}{2}$ $\frac{x_2}{4}$ $\frac{x_3}{8}$

Apply controlled $R_3$ from qubit $3 \to 1$

$$R_3 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^3}} \end{pmatrix}$$




$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_1x_2}|1\rangle\right)|x_2\cdots x_n\rangle$$

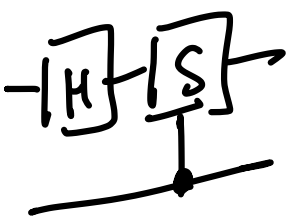
$$\to \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle\right)|x_2\cdots x_n\rangle$$

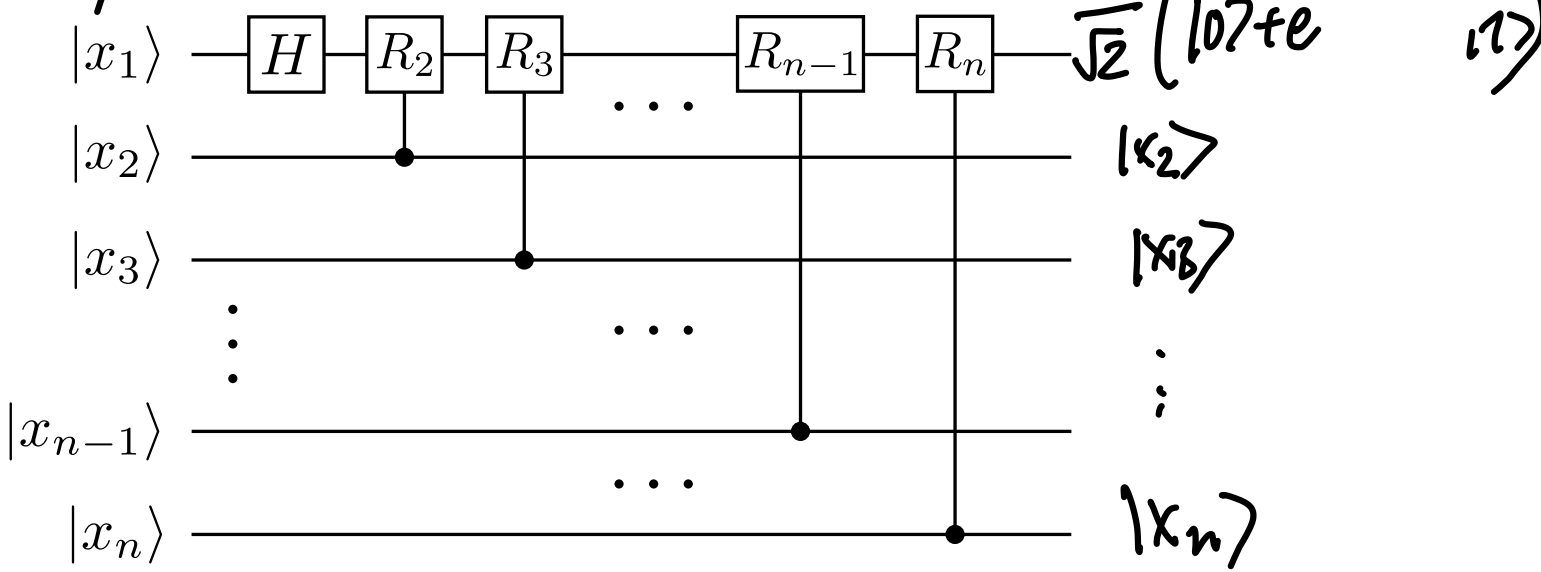
# A circuit for the QFT

$$|x\rangle \to \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

Apply a controlled $R_4$ from $4 \to 1$, etc. up to the $n$-th qubit to get

$\begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{pmatrix}$ $k=2 \Rightarrow S$ gate

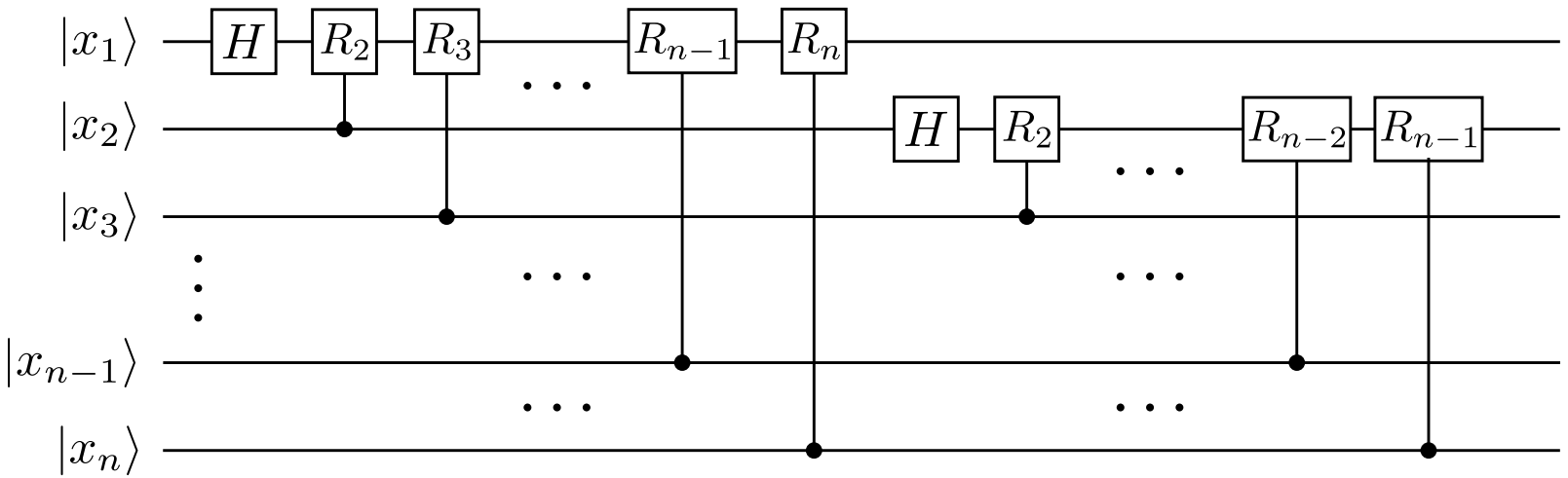


# A circuit for the QFT

Similar thing with second qubit: apply $H$ then controlled rotations from qubits 3 to $n$ to get

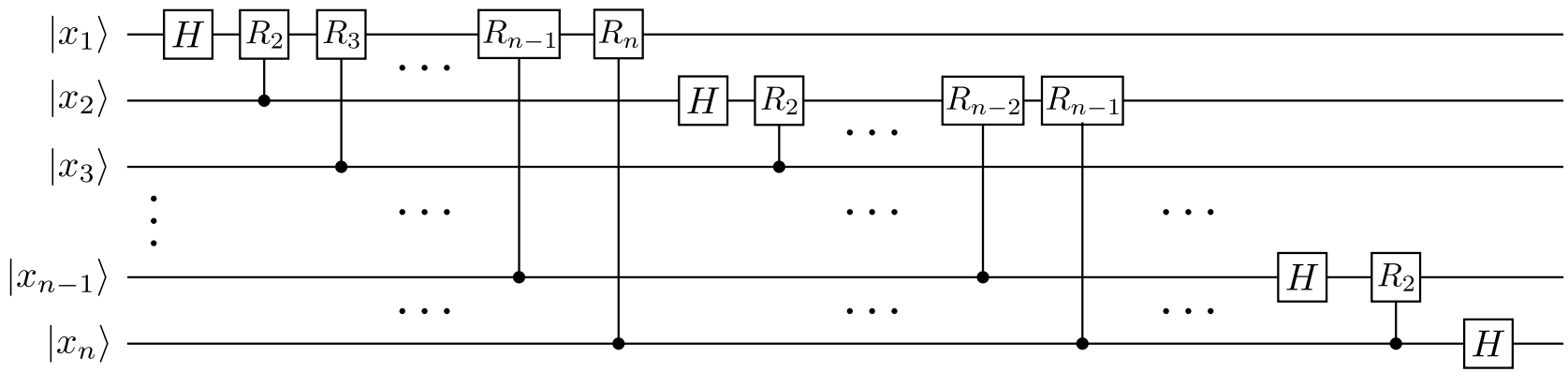
$$\left(|0\rangle + e^{2\pi i\, 0.x_1 \cdots x_n}|1\rangle\right) \otimes \left(|0\rangle + e^{2\pi i\, 0.x_2 \cdots x_n}|1\rangle\right) \otimes |x_3 \cdots x_n\rangle$$



# A circuit for the QFT

Do this for all qubits to get that big ugly state from earlier:

$$|x\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$




Qubit order is backwards - easily fixed with SWAP gates.

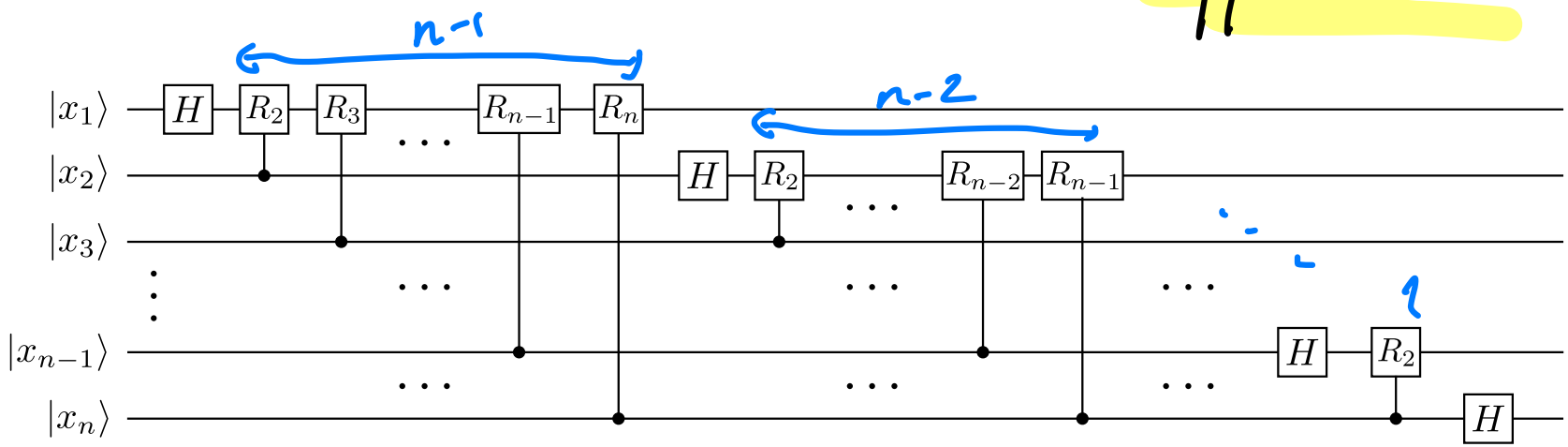
# Quantum Fourier transform

(as a func. of $n$)

**Exercise**: What are the gate counts and depth of this circuit?

- $H$ : $n$
- controlled rotations : $\frac{n(n-1)}{2}$ $(O(n^2))$
- SWAP count (if you care about order) : $\lfloor n/2 \rfloor$
- depth : $2n$

qp. QFT



# Next time

Content:

- Toward's Shor's algorithm: order finding with QPE

Action items:

1. Finish tutorial assignment 2
2. Work on project (and fill peer assessment)

Recommended reading:

- From today: Codebook QFT, QPE, Nielsen & Chuang 5.1-2
- For next class: Codebook SH, Nielsen & Chuang 5.3.1

# Derivation of output state of QFT

We will reexpress $k/N$ in fractional binary notation, then reshuffle and *factor* the output state.

$$
\begin{aligned}
|x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x(k/N)} |k\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} e^{2\pi i x\left(\sum_{\ell=1}^{n} k_\ell 2^{-\ell}\right)} |k_1 \cdots k_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \bigotimes_{\ell=1}^{n} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle
\end{aligned}
$$

# Derivation of output state of QFT

$$
\begin{aligned}
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \bigotimes_{\ell=1}^{n} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \\
&= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^{n} \left( \sum_{k_\ell=0}^{1} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \right) \\
&= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^{n} \left( |0\rangle + e^{2\pi i x 2^{-\ell}} |1\rangle \right) \\
&= \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right) \cdots \left(|0\rangle + e^{2\pi i 0.x_1 \cdots x_n}|1\rangle\right)}{\sqrt{N}}
\end{aligned}
$$

# Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let $\lambda_k$ be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary $U$:

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\begin{aligned}\langle k| U^\dagger U|k\rangle &= \langle k| \lambda_k^*\lambda_k|k\rangle \\ \langle k|k\rangle &= |\lambda_k|^2\langle k|k\rangle \\ 1 &= |\lambda_k|^2\end{aligned}$$